Item 7: the cubic-GHS to genus-3 transfer, priced end to end against rho

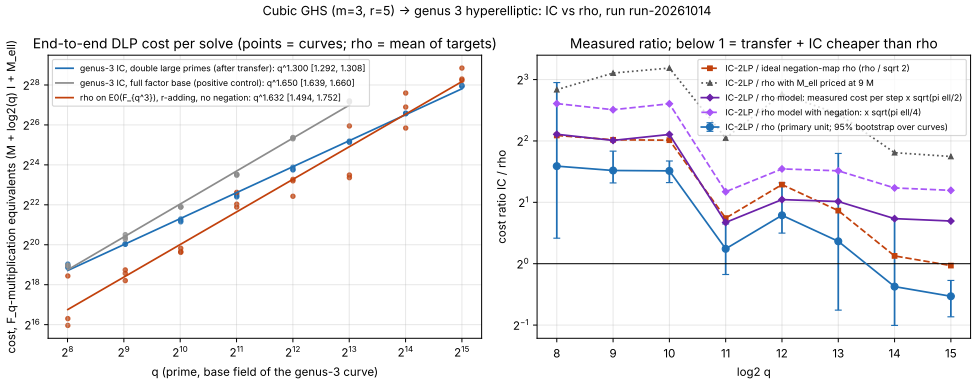
2026-10-14. Campaign `research/ic_next10_20261014/` (rules: ../RULES.md). Design: `reports/Index calculus leads next steps.md`, "### 7.". Protocol, frozen before data, with amendment A1 (before recorded data) and deviations D1-D2 (after data): PROTOCOL.md. Tool: ghs3.py (Python 3, single-threaded, explicit operation counters). Analysis and plot: analyze.py. Records: results/run-20261014/.

Verdict

Hyperelliptic branch (`m = 3, r = 5`): POSITIVE under the frozen decision rule, at toy scale, with a qualification on the crossover. Plane-quartic branch (`m = 2`, Diem): BLOCKED (no class arithmetic on the plane quartic in the repository; not attempted).

- Complete DLP solves through the transfer: **24/24 correct** for genus-3 IC with double large primes (q = 2^8 to 2^15, three curves per size), 18/18 for the full-factor-base control (q <= 2^13), 66/66 for rho on `E0(F_{q^3})`, and 72/72 for the random-multiplier rho control. Every answer is checked against the planted `k`.
- Fitted cost exponent in `q` (primary unit, all sizes, 95% bootstrap over curves): **IC-2LP 1.300 [1.292, 1.308]**; positive control IC-basic 1.650 [1.639, 1.660]; rho, measured, 1.632 [1.494, 1.752]; rho model (measured cost per step times `sqrt(pi ell / 2)`) 1.529.
- **The 1.35 kill line is crossed if only the larger sizes are fitted.** The IC fit over q = 2^11 to 2^15 gives **1.377 [1.363, 1.392]**, and the local exponents between consecutive sizes rise: 1.08, 1.20, 1.25, 1.35, 1.35, 1.42, 1.39. The all-size fit is pulled down by fixed costs (transfer, walk set-up, factor-base scan) that dominate at q = 2^8 to 2^10. Both fits stay separated from rho's 1.5 (their intervals end at 1.39).
- **Measured crossover:** against the rho arm as implemented, IC-2LP is cheaper from **q = 2^14** (ratio 0.77 [0.50, 1.65]) and at **q = 2^15** (0.69 [0.55, 0.83]). That rho arm walks badly (1.75 sqrt(ell) steps against 1.35 sqrt(ell) for independent random multipliers, deviation D2). **Against the strongest rho the instrument supports (rho model), IC-2LP is still 1.62 times more expensive at q = 2^15**, and 2.29 times against an ideal negation-map rho. The crossover against the rho model is an extrapolation of fitted lines: q = 2^17.2 (all-size IC fit) to 2^19.2 (top-five-size fit); against negation-map rho, 2^19.4 to 2^22.5; with linear-algebra multiplications mod `ell` priced at 9 `F_q` multiplications, about 2^25. No crossover against the rho model is measured in this run.

What the result supports: on this structured odd-characteristic class (`#E0(F_{q^3}) = 0 mod 4`, split arm), the transfer plus genus-3 double-large-prime IC runs end to end and scales with an exponent between 1.30 and 1.39 in `q`, against 1.5 for rho on `E0`. It does not support a claim that the transfer beats a well-implemented rho at any size measured here.



cost against q and IC/rho ratio

Left: end-to-end cost per solve in `F_q`-multiplication equivalents (points are curves; rho points are means over 2-3 targets; lines are the least-squares fits with 95% bootstrap intervals). Right: IC-2LP / rho per size, against the measured rho (with bootstrap interval), the rho model with and without negation, and with `M_ell = 9 M`. Run `run-20261014`, 3 curves per size, `n` per cell in the table below.

Per-size results (means; costs in `F_q`-multiplication equivalents)

log2 q	q	IC-2LP cost	IC-basic cost	rho cost (measured)	rho model	IC/rho measured [95%]	IC/rho model	IC/rho model, negation	IC LA share	FB size	IC walk steps	correct (IC, basic, rho)
8	257	5.01e5	5.09e5	1.67e5	1.16e5	3.01 [1.33, 7.73]	4.31	6.10	0.17	81	914	3/3, 3/3, 9/9
9	521	1.08e6	1.37e6	3.76e5	2.68e5	2.87 [2.49, 3.57]	4.02	5.69	0.25	130	1,755	3/3, 3/3, 9/9
10	1031	2.43e6	3.90e6	8.53e5	5.66e5	2.85 [2.50, 3.19]	4.30	6.09	0.27	205	3,754	3/3, 3/3, 9/9
11	2053	5.77e6	1.19e7	4.87e6	3.62e6	1.18 [0.89, 1.57]	1.59	2.25	0.31	324	8,285	3/3, 3/3, 9/9
12	4099	1.46e7	4.26e7	8.47e6	7.09e6	1.73 [1.41, 2.49]	2.06	2.92	0.37	513	18,800	3/3, 3/3, 9/9
13	8209	3.75e7	1.52e8	2.91e7	1.86e7	1.29 [0.59, 3.47]	2.02	2.85	0.40	814	43,445	3/3, 3/3, 9/9
14	16411	9.99e7	-	1.29e8	6.00e7	0.77 [0.50, 1.65]	1.66	2.35	0.44	1292	101,978	3/3, -, 9/9
15	32771	2.60e8	-	3.76e8	1.61e8	0.69 [0.55, 0.83]	1.62	2.29	0.48	2049	243,240	3/3, -, 6/6

Rho cost depends on `ell = #E0(K)/c`, and the cofactor `c` varies from 4 to 60 across curves (instances.csv), while IC cost does not depend on `c`. That is why the ratio jumps at q = 2^11, where two of three curves have `c = 4`. Every comparison is per instance on the same `ell`-part.

Other measured values (from summary.json):

- Rho steps scale as `ell^0.527` (instrument control; expected 0.5). Measured rho constant: 1.97 sqrt(ell) over all sizes, 1.75 at q = 2^9 to 2^12. With

independent random multipliers at the same sizes: 1.35 +- 0.08 sqrt(ell) (n = 72). Random mapping predicts 1.25.
- Rho cost per step: 74.5 `F_q`-multiplication equivalents (affine addition over `F_{q^3}`: one `F_{q^3}` inversion by norm, three `F_{q^3}` multiplications by schoolbook).
- IC-2LP group operations scale as q^1.137; its cost exponent exceeds that because the smoothness test costs `log q` and the linear-algebra share grows from 17% to 48% (Lanczos on the normal equations, row weight from large-prime recombination rising slowly with `q`).
- At q = 2^15 an IC-2LP solve takes about 243,000 walk steps, 2,049 factor-base primes, 81-95 full relations, about 1,600 single- and 11,500-11,700 double-large-prime relations, and a single Lanczos solve; walk restarts at q >= 2^13: one in nine solves.
- All 24 instances pass the transfer checks: `deg h = 7`; `phi(G) != 0`; `ell phi(G) = 0`; `#E0(K) phi(G) = 0`; 3/3 random homomorphism checks `phi(P+Q) = phi(P) + phi(Q)`. At q <= 2^11, `#Jac(C)(F_q)` from PARI `hyperellcharpoly` equals `#E0(K)` from SEA on 12/12.

What was built

1. **Instances.** `K = F_q[t]/(t^3 - c1 t - c0)`, `q` the next prime above 2^b. `E0 : y^2 = (x-a)(x-b)(x-r0)(x-r1)`, `r_i = p^{q^i}`, `p` random in `K \ F_q`; accepted when `#E0(K) = c ell` with `ell` prime, `c <= 64`.
2. **Genus-3 curve over `F_q`.** `D = C/iota` is the conic `Tr_{K/F_q}((r2 - r0) s^2) = 0` in `s in K`, with `X_i = s^{q^i}`. Lines through an `F_q`-point `s*` parametrise it; the second line is chosen through the `F_q`-point over `x = a`, so the Weierstrass point sits at `z = inf` and `C : w^2 = h(z) = (xn - a xd)(xn - b xd)` has degree 7.
3. **Transfer, computed over `K` only.** With `R(z) = kappa Delta^2 / ((r0 - r1) X0 X2)` (derived: `w = y0 y1 y2 xd/(Q N) = y0 R(z)`), `pi0^*[P]` is the Mumford pair `(xn - x_P xd, y_P R mod U)`; `phi(P)` is the Cantor sum of it and its two coefficient-wise Frobenius conjugates, and lands in `F_q`. This replaces the `F_{q^6}` Zech tables of the earlier build, which limited it to q <= 13.
4. **IC-2LP.** r-adding walk (r = 20) on `alpha phi(G) + beta phi(T)`; smoothness by `x^q = x mod u`; roots by Cantor-Zassenhaus; factor base the `ceil(2 q^{2/3})` degree-1 primes with smallest `x` (kappa = 2 from the pilot); large primes eliminated along the spanning tree of the component of the large-prime graph that contains the factor-base vertex; singleton filtering; Lanczos mod `ell` with `k` as an extra unknown; answer checked as `k phi(G) = phi(T)` and against the planted `k`.
5. **Rho.** Single-walk r-adding rho with distinguished points on the short Weierstrass model of `E0` over `K`.

Deviations and limits

- Sizes extended from 2^8 to 2^13 to 2^8 to 2^15 (A1, before recorded data); stopped at 2^15 because a 2^16 rho job would hit the 45-minute per-job cap (D1). The kill criterion's "q >= 2^9" reach condition is met at every size.
- Rho multipliers by a Fibonacci chain walk about 30% worse than random multipliers (D2); the verdict uses the rho model, which removes that

handicap.

- One instrument on all arms, but not equally tuned: Cantor's generic composition (no explicit genus-3 formulas), schoolbook `F_{q^3}` multiplication (no Karatsuba), Fermat-priced inversions. Explicit genus-3 formulas would lower IC's constant; Karatsuba and a negation map would lower rho's. The crossover moves by a factor `ratio^{6}` in `q` for a constant change `ratio` (exponent gap about 1/6 to 1/4), so it is sensitive to these constants: a 1.6x constant is 2^3 to 2^4 in `q`.
- `M_ell` counted as one `F_q` multiplication (one machine word at these sizes). With 9, the IC exponent is 1.438 and the crossover moves to about 2^25.
- Python stands in for native code because the task allows it; costs are counted operations, not wall time.
- Only the split arm (`deg h = 7`) is covered; the conjugate arm (`deg h = 8`, real model) and the plane-quartic branch are not.

Reproduce

```
        cd research/ic_next10_20261014/07-cubic-ghs-genus3
        # kappa pilot (q = 257)
        for k in 1 2 4; do for c in 0 1 2; do python3 -I ghs3.py --
bits 8 --curve $c --arms ic2lp --kappa $k --out
results/pilot_kappa_k$k.jsonl; done; done
        # recorded run (b = 8..15; PAR = parallel jobs; each job
single-threaded, 1 GB cap)
        PAR=3 sh run_all.sh results/run-20261014 8 9 10 11 12 13 14 15
        # random-multiplier rho control (deviation D2)
        for b in 9 10 11 12; do for c in 0 1 2; do RHO_RANDOM_MULT=1
python3 -I ghs3.py --bits $b --curve $c --arms rho --rho-targets 6 --
out results/rho-randmult-20261014/rho_b${b}_c${c}.jsonl; done; done
        python3 -I analyze.py results/run-20261014
```

Toolchain and hashes: results/run-20261014/env.txt (`ghs3.py` later gained only the `RHO_RANDOM_MULT` switch, default off). Per-solve rows: solves.csv.